CHINMAY MHATRE

(412) 478-6216 ◇ Pittsburgh, PA
mhatrec82@gmail.com Google Scholar Chinmay Mhatre

EXPERIENCE

Research Associate - Computational Chemistry July 2026 – Present
University of Pittsburgh *Sponsor: Naval Research Laboratory*

- Built an end-to-end Python pipeline for high-throughput electronic structure analysis, parsing Gaussian output at scale to extract dipole moments, Mayer and Wiberg bond orders, and condensed-to-atoms density matrices; exported to sharded HDF5 for downstream ML training on QM9-scale datasets.

Doctoral Researcher Aug 2020 – May 2026
Swanson School of Engineering, University of Pittsburgh, Pittsburgh, PA
Thesis: Computational investigation of atomistic phenomena in Zr-based Metal-Organic Frameworks and stainless steel using classical and quantum mechanical calculations

Machine Learning Interatomic Potentials for Reactive Systems *Funded by DTRA*

- To resolve catalyst performance loss that DFT could not explain at scale, deployed a trained AI/ML interatomic potential across 100+ reactive molecular dynamics simulations of catalytic surface kinetics, diagnosing the surface poisoning mechanism and delivering redesign guidance to experimental partners.
- To reduce the high cost of ML training-data generation, benchmarked the RAL protocol against the published standard protocol, building the Python training pipeline for the standard on HPC, cutting training cost 50% while designing a latent-space uncertainty metric for auditable, reproducible model deployment.
- To extend reactive modeling to a dual-function system, executing the Reactive Active Learning protocol for chemical threat neutralization reactions in MOF-808 (ongoing DTRA-funded work).

Molecular Transport in Defective UiO-66 *Funded by DTRA*

- To overcome slow molecular transport limiting the material's performance, engineered ordered point defects in UiO-66 and achieved a 100-fold improvement in isopropanol diffusivity over a randomly defective structure.
- To identify the structural factors controlling transport, quantified self- and transport diffusivity of polar IPA across four concentrations using 10 independent MSD analyses per datapoint (LAMMPS) and GCMC (RASPA), demonstrating concentration-dependent suppression in hydroxylated variants.
- To determine what controls reactivity, mapped hydrolysis reaction pathways across four capping-group chemistries with CI-NEB (VASP) and established the binding mechanism as the controlling factor in hydrolysis rates.
- To enable experimental validation, generated DFT-derived IR spectra providing vibrational fingerprints for capping-group-modified MOFs.

Reactive Force Field Development for Stainless Steel Corrosion *Funded by Naval Research Laboratory*

- To model chloride-induced pitting corrosion at the atomic scale, developing and validating Cr/O/Cl ReaxFF parameters in LAMMPS for chromium oxide passive films (ongoing NRL-funded work).

Thermophysical Properties of Fuel Additives *Funded by Lubrizol Corporation*

- To give Lubrizol a baseline for molecules with no experimental data, provided the first-ever critical temperature and pressure estimates for three organometallics and their precursors via Gibbs ensemble Monte Carlo (Cassandra), validated within 10% of available experimental densities.

Graduate Student Intern May 2023 – Nov 2023
Chemical Separations Group, Oak Ridge National Laboratory, Oak Ridge, TN

- To give a DoE purification program a validated computational standard, benchmarked six free-energy solvation workflows against experimental partition coefficients and delivered the most accurate protocol as the adopted standard.
- To accelerate the experimental design cycle, conducted SAR analysis across 10+ molecular modifications and identified the top-performing structural motif using the benchmarked protocol.

PUBLICATIONS

Mhatre, C. V., Wardzala, J. J., Shukla, P. B., Agrawal, M., Johnson, J. K. (2023). Calculation of self, corrected, and transport diffusivities of isopropyl alcohol in UiO-66. *Nanomaterials*, 13(11), 1793. DOI:10.3390/nano13111793

Mhatre, C. V., Wardzala, J. J., Oliver, M. C., Islamov, M., Boone, P., Wilmer, C., Huang, L., Johnson, J. K. (2024). The Impact of Missing Linker Defects in UiO-66 on Adsorption and Diffusion of Isopropyl Alcohol. *The Journal of Physical Chemistry C*, 128(32), 13577–13587. DOI:10.1021/acs.jpcc.4c02982

Achar, S. K., Shukla, P. B., Mhatre, C. V., Bernasconi, L., Vinger, C. Y., Johnson, J. K. (2025). Reactive Active Learning: An Efficient Approach for Training Reaction Aware Machine Learning Potentials. *Journal of Chemical Theory and Computation*, 21, 18, 8889–8906. DOI:10.1021/acs.jctc.5c00920

Johnson, J. K., Kin Mai Lai, S., Mhatre, C. V., Williams, S., Keith, J. A., Call, M., Cormack, G. Estimation of Dipole Moments of Zinc Dialkyldithiophosphates and Their Precursors. (*under review*)

Mhatre, C. V., Gibbons, B., Morris, A. J., Johnson, K. Effect of Capping Group Binding on Chemical Reactivity. (*in preparation*)

Mhatre C.V., Zulueta B., Keith J. Representing large chemical structure datasets for Quantum and semiempirical methods comparison (in preparation)

Mhatre, C. V., Van Duin, A., Keith, J. A. Developing Cr/O/Cl ReaxFF Parameters to Understand Stainless Steel Depassivation in Saline Environments. (*in preparation*)

Mhatre, C. V., Gordon, A. Y., Ananthabhotala, L. Y., Achar, S., Johnson, J. K. Reactions of GB and HD in MOF-808-NH_2 Predicted from Machine Learning Interatomic Potentials. (*in preparation*)

EDUCATION

Ph.D., Chemical Engineering Aug 2020 - May 2026
University of Pittsburgh, Pittsburgh, USA
Advisors: Prof. J. Karl Johnson and Prof. John Keith

Bachelor of Chemical Engineering Aug 2016 – Aug 2020
Institute of Chemical Technology, Mumbai, India

SKILLS

Simulation	DFT, MD (metadynamics, MSD), MLIP development, GCMC, Gibbs ensemble MC, ReaxFF parametrization, vibrational frequency analysis
Software	LAMMPS, VASP, PLUMED, WHAM, PACKMOL, Gaussian, Orca, Quantum Espresso OpenMM, openbabel, CREST, pymol, RASPA
Machine Learning	PyTorch, TensorFlow, DeepMD (active learning), MACE, UMA, scikit-learn, CNN, GAN, VAE, equivariant neural networks, WandB, Google Cloud, AWS
Programming	Python (numpy, pandas, scipy, ASE, RDKit, UMAP, pymatgen, catkit, multiprocessing), Linux/Bash, MATLAB, SQL, Git, HPC (Slurm, CPU, GPU), LaTeX
Data & Reports	Microsoft Office Suite, Power BI, DB Browser

SELECTED PRESENTATIONS (7 OF 11)

Achar, S., Mhatre, C., et al. (2025, Nov). Computational Design and ML Assisted Search of Dual Function MOFs Towards Chemical Warfare Agents' Decomposition. *AIChE Annual Meeting.*

Achar, S., Mhatre, C., et al. (2025, June). CH Bond Activation on Transition Metal Carbides: An MLFF Approach with Reactive Active Learning. *29th North American Catalysis Society Meeting.*

Mhatre, C., Policastro, S., Anderson, R., van Duin, A., Keith, J. (2024, Oct). Reactive Force Field Parametrizations to Examine Atomic Scale Processes on Stainless Steel in Chloride Environments. *AIChE Annual Meeting.*

Mhatre, C., Wardzala, J., et al. (2023, Nov). Diffusion and Adsorption of Polar Molecules in Defective UiO-66. *AIChE Annual Meeting.*

Mhatre, C., Gibbons, B., Morris, A. J., Johnson, K. (2023, Nov). Effect of Capping Group Binding on Chemical Reactivity. *AIChE Annual Meeting.*

Mhatre, C., Wardzala, J., et al. (2022, Nov). Modeling of Adsorption and Diffusion of Chemical Agents and Simulants in UiO-66. *AIChE Annual Meeting.*

Wardzala, J., Shukla, P. B., Mhatre, C., Johnson, K. (2021, Nov). Modeling of Diffusion in the Metal-Organic Framework UiO-66. *AIChE Annual Meeting.*

LEADERSHIP

Board Member, Engineering Graduate Student Organization (EGSO), University of Pittsburgh 2022 – 2025
Roles: Vice President, Secretary, Professional Development Officer
Coordinated graduate student events across the School of Engineering; managed a team of eight in planning and logistics for 500+ engineering graduates.

Teaching Assistant, University of Pittsburgh 2022 – 2024
Chemical Reaction Engineering, Chemical Engineering Thermodynamics, Mathematical Methods (avg. class size: 50). Developed course material and capstone projects.

AWARDS AND CERTIFICATIONS

- Pharmaceutical Formulations, Schrödinger Inc. 2026
- Introduction to Large Language Models, LinkedIn Learning 2025
- Deep Learning with PyTorch: Generative Adversarial Networks, Coursera 2025
- DTRA Grant: Computational design and ML-assisted search for dual-function MOFs towards nerve and blistering agent detoxification 2023
- ACS Reviewer Lab 2021
- Ongoing:
  - Databases and SQL for Data Science with Python, IBM/Coursera
  - Statistical Thinking for Industrial Problem Solving, JMP/Coursera
  - Building AI agents and Agentic Workflows, IBM/Coursera
  - Applied Bayesian Data Analytics, University of Pittsburgh/Coursera